ANNEXURE - II

FORMAT - TECHNICAL SPECIFICATION

For Fully automatic TANUVAS GRAND Production plant
(Capacity – 15,000 litres / day or 25,000 bottles / day @ 600ml / bottle) with all accessories

Note: Kindly adhere to the technical specification of your firm as per the technical specifications mentioned in annexure I. If you are adhering to the technical specification mentioned in annexure I, then kindly mention as "As per requirement" in column III. If any deviations are made, mention the deviations very clearly in column III.

S.No.	Description (Column I)	Required specification (Column II)	Mention your firm Specifications without deviations for all parameters. Mention specifically if any deviations made (Column III)
1.0	**FULLY AUTOMATIC TANUVAS GRAND MIXER (3000 LITRES CAPACITY) AND STORAGE TANK (3000 LITRES CAPACITY) WITH ACCESSORIES INCLUDING ELECTRIC PANEL**		
1.01	**TANUVAS GRAND MIXING VESSEL – 3000 LTR**		
a	**GENERAL**		
	Material Of Construction	SS 316 contact parts and SS 304 non contact parts.	
	Vessel Type	Cylindrical Vertical with welded top / bottom dish end	
b	**PROCESS DETAILS& OPERATIONAL DATA**		
	Specific Gravity	0.92gms /cc	
	Operating Temperature	Ambient	
c	**DESIGN DATA**		
	Working Volume	3000 Ltr.	
	Gross Volume.	3600 Ltr	
d	**MATERIAL OF CONSTRUCTION**		
	Shell / Bottom / Top dish	Stainless Steel 316 Grade	

e	**DRIVE AND AGITATOR DATA**		
	Type of agitator	Baby Anchor with 2 Blade PBT	
	Mixing speed	60 minutes for 3000 litres	
	Motor HP Main Drive	15.0 HP/1440 RPM / Flange Mounted or equivalent	
f	**NOZZLES - TOP DISH**		
	Man Hole	450 mm with dished cover and 100 mm sight glass	
	Light Glass	100 MM Pad Type with FLP Vessel lamp	
g	**SHELL**		
	Temperature Sensor	½" BSP	
	Phase Inlet	40 mm with valve	
	Recirculation	50 mm with Valve	
h	**BOTTOM DISH**		
	Product Outlet	50 MM with ball valve	
	Filter	Suitable filter (<200 micron)	
i	**SURFACE FINISH**		
	Product Contact Parts	Polished to mirror finish. Surface roughness < 0.5 Ra	
	Product non contact parts	Polished to matt finish. Surface roughness < 0.8 Ra	
1.02	**TANUVAS GRAND STORAGE VESSEL – 3000 LTR**		
a	**GENERAL**		
	MOC	SS 316 contact parts and SS 304 non contact parts.	

	Vessel Type	Cylindrical Vertical with welded top / bottom dish end	
b	**Process and Operation Data**		
	Gross Volume (L)	3300 Ltrs	
	Working Volume (L)	3000Ltrs	
	Nozzles	Dia450mm, ManHole 50 mmInlet, 40 mm Spare25 mm outlet with valve	
c	**DESIGN DATA**		
	Main Shell ID	ID Dia. 1575 mm	
	Height of Shell / Cone	1500 / 500	
d	**AGITATOR DATA (SUBJECT TO REVALIDATION OF DESIGN DATA)**		
	Type of agitator	Baby Anchor with 2 Blade PBT	
	Agitator Entry	Top Entry	
	Motor HP Main Drive	15.0 HP/1440 RPM / Flange Mounted or equivalent	
e	**MATERIAL OF CONSTRUCTION**		
	Product Contact Parts	Stainless Steel 316 Grade	
	Product non contact parts	Stainless Steel 304 Grade	
f	**BOTTOM DISH**		
	Product Outlet	50 MM with ball valve	
	Filter	Suitable filter (<200 micron)	
g	**SURFACE FINISH**		
	Product Contact Parts	Polished to mirror finish. Surface roughness < 0.5 Ra	

	Product non contact parts	Polished to matt finish. Surface roughness < 0.8 Ra	
2.0	**FULLY AUTOMATIC TANUVAS GRAND LIQUID FILLING IN BOTTLE AND INNER PLUGGING AND OUTER CAPPING (60 BOTTLES PER MINUTE) WITH ACCESSORIES INCLUDING ELECTRIC PANEL FOR EACH PART**		
2.01	**TURN TABLE/BOTTLE FEEDER/CONTAINER LOADER**		
a	**Operation Requirements**		
	Container inputs in turn table by manually and automatically will rotate on disk of turn table and exit through outlet, during rotation of Container will guided by a SS strip, which will guide the container towards outlet path.		
b	**Technical Specifications**		
	Machine type	Fully automatic turn	
	Direction	Clockwise / Anti clock wise as per requirement	
	Power and Motor	0.5 to 1.0 HP/220 to 415 V, 50 to 100 cycles/ second or equivalent	
	Output	100 to 120 Product /Minute (Required speed in filling line, 60BPM)	
c	**Salient Features** All exposed parts are of S.S. 304 to prevent rusting Required conveyor from auto turn table to air-jet cleaning machine to maintain the speed.		
2.02	**FULLY AUTOMATIC VERTICAL AIR-JET CLEANING MACHINE**		
a	**Technical Specifications**		
	Machine type	Fully Automatic	
	Out put	60 Bottles per Minute	

	Motor Specification	0.5 HP / 415 Volts / 50 Hz. 0.5 HP x 2 = 2 HP / 415 Volts / 50 Hz. or equivalent	
	Cleaning Cycle	Positive air pressure and vacuum cleaning	
2.03	**FULLY AUTOMATIC VOLUMETRIC LIQUID FILLING MACHINE IN BOTTLES**		
	All parts should be made from **SS-316** Materials		
a	**GRAND liquid Bottle size for reference**		
	Bottle size – Height 205mm, Diameter – 235mm, Neck size – 80mm.		
b	**Operation Requirements**		
	Nozzle should go upwards slowly from the bottom level of bottle towards neck during filling to minimize foaming adjustable nozzle is reciprocating according to filling dose.		
c	**Technical Specifications**		
	Machine	Fully automatic	
	Output	Up to 60 Bottles per minute	
	Filling Volume	600ml	
	Electrical Specification		
	Main Machine	2.5 HP / 415 Volts / 50 Hz or equivalent for output of 60BPM	
	Conveyor	1.5 HP / 415 Volts / 50 Hz or equivalent for output of 60BPM	
	Storage tank	SS316 with the capacity of more than 500 litres	
	SS conveyor belt made from the SS316 (Length as per the speed of bottles filling – 60BPM) Adjustable Conveyor height as per choice to align with another line machine.		

2.04	**FULLY AUTOMATIC INNER / OUTER PLUGGING MACHINE**		
	Suitable to capping inner caps output - 60 Bottles/Minute (Bottle outer cap size: Height – above 25mm, Diameter – Above 100mm). One complete set of change parts suitable for any one size diameter of caps & bottle. Total machine construction in SS-304 wherever; Contact parts - SS-316 material.		
a	**Operation Requirements**		
	Outer cap should be placed according to the 60 bottles filling speed.		
b	**Technical Specifications**		
	Machine output	Up to 60 Bottles per minute	
	MOC of machine frame	Mild Steel powder coated and outside cladding SS 304	
	MOC of Contact Parts	Stainless Steel 316	
	Machine Covers	Stainless Steel 304	
	MOC of all vital shafts	EN 31 / Stainless Steel 304/316	
	MOC of Conveyor	Conveyor Slat Stainless Steel 304 Slat with Nylon Strips on Side Conveyor Shafts Mild Steel with Hard Chrome or Zinc Plating Conveyor Sprockets Mild Steel or Nylon	
2.05	**FULLY AUTOMATIC OUTER CAPPING MACHINE**		
	Suitable to capping outer caps output - 60 Bottles/Minute (Bottle inner cap size: Height – 23mm and Diameter – 100mm). One complete set of change parts suitable for any one size diameter of caps &bottle. Total machine construction in SS-304 wherever; Contact parts - SS-316 material.		

a	**Technical Specification**		
	Output	Upto30-60BottlesperMinute	
	No. of Sealing Head	According to the output of 60 BPM	
	Capping Unit	Magnetic Screw Capping Head with Torque Adjustment	
	Caps Feeding	Vibratory Bowl Feeder	
	Caps Hammering	Pneumatic System	
	Conveyor Length	Depending upon the product output of 60BPM	
b	**Machine's material of construction**		
	Machine main frame	Square Pipe, Stainless Steel 304	
	Machine Coverings	Complete in Stainless Steel 304	
	Product Contact Parts	Stainless Steel 316	
c	**Machine's electronics & pneumatics components**		
	Capping Head Motor Make	1 H.P Standard Model	
	Capping Head VFD Make	1 H.P Delta	
d	**Machine's power & air consumption requirement**		
	Power	1 H.P / 230 V/AC or 1.5KW or equivalent	
	Phase	Single Phase/ Three Phase according to the requirement	
	Compressed Air Required	3 H.P Compressor (6 bar pressure 0.5 CFM) or equivalent	

3.0	**TANUVAS GRAND LIQUID BOTTLES DRYING TUNNEL (60 BOTTLES PER MINUTE) WITH ACCESSORIES INCLUDING ELECTRIC PANEL FOR EACH PART** The fully automatic drying tunnel should be designed with precision, these machines are widely recognized for their durability, design excellence and high performance.		
a	**Technical Specifications**		
	Machine Type	Fully automatic	
	Temp. Req.	100’ To 150’ Degree Centigrade in Chamber,	
	Heater	Heater With Heating System. Total 10 To 24 Kw	
	Motive Load	1.5 Hp Conveyor and 0.5 Hp Blower Fan Load	
b	**Material**		
	•Incoming MCB 32amp indo copper make or equivalent. •Blower Motor’s are remi make or equivalent. •Blower Motor are separately controlled.		
4.0	**FULLY AUTOMATIC TANUVAS GRAND LIQUID BOTTLES PACKING IN CARTON BOX UNIT WITH ACCESSORIES FOR INSTALLATION, PACKING, CLOSING AND STRAPPING FOR TRANSPORTATION – REQUIRED NO. 1 SET**		
	Carton Box size for reference: Length - 470mm, Width – 390mm and Height – 270mm		
a	***S-Conveyor***		
b	**Technical Specifications**		
	Output per Hour	60 to 65 Bottle/Minute	

	Working Desk Height	860 To 910 mm according to the requirement	
	Power supply	240V /15A/50Hz/-1Ph according to the requirement	
	Power consumption	0.75 Kw according to the requirement	
	Speed	0-11 mtr/min	
c	**Case Erector**		
d	**Technical Specifications**		
	Output per Hour (No. of cases)	8-10/Minute	
	Power supply	240V /15A/50Hz/-1Ph according to the requirement	
	Power consumption	0.75 Kw according to the requirement	
	Speed	0-10 mtr/min	
	Material of construction	MS powder coated	
e	***Roller Conveyor***		
	Power supply	240V /15A/50Hz/-1Ph according to the requirement	
	Power consumption	0.37 Kw according to the requirement	
	Speed	0-20 mtr/min	
	Drive roller	SS-304	
f	***Case Tapping Machine***		

g	**Technical Specifications**		
	Power supply	240V /15A/50Hz/-1Phaccording to the requirement	
	Power consumption	0.75 Kwaccording to the requirement	
	Speed	0-20 mtr/min	
h	***Pick and Place Unit***		
	Power supply	240V /15A/50Hz/-1Phaccording to the requirement	
	Power consumption	0.37 Kwaccording to the requirement	
	Speed	0-20 mtr/min	
i	***Idler Roller Conveyor***		
	After case sealing from top of the box filled box will come and accumulate on this gravity roller conveyor. From this point filled box will pick manually		
j	***Carton box strapping machine***		
k	**Technical Specifications**		
	Output per Hour (No. of cases)	2-5/Minute or equal	
	Direction of Movement	Left to Right	
	Power supply	100, 110, 220, 230, 240 V., 50 / 60 Hz according to the requirement	
	Power consumption	400 watt	

5.0	**FULLY AUTOMATIC BOTTLE PRINTER TO PRINT PLACE OF MANUFACTURE, DATE OF MANUFACTURE, BATCH CODE AND EXPIRY DATE**		
a	**Technical specifications**		
	Printing speed	Intermittent 100 mm/s,150 mm/s,200 mm/s, 225 mm/s, Continuous up to 450 mm/s	
	Voltage	90 V to 260 V,47 Hz to 63 Hz	
	Resolution	8 dots per mm	
	Print area	30mm to 50mm	
	Message Storage Capability	250	
	Maximum Lines of Print	5, 32pixel	
	Printing speed	Intermittent 100 mm/s,150 mm/s,200 mm/s, 225 mm/s, Continuous up to 450 mm/s	